STARFISH SPACE

TVAC System

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About Me

- I’m studying Mechanical Engineering (Course II) at MIT (shoutout Cappie & Sean !)
- I grew up near Atlanta, GA
- Goals?
  - I want to work on spacecraft in some capacity (sats, rockets, landers, who knows…)
- Proud shrimp gang  
- This is my first industry experience – My work felt important and the people are incredible!!!

From MoPop

Starfish Summer Wrapped

- ~ 198 trains/buses taken to work/Seattle/hikes
- 144 km ran
- 61 km hiked
- 20 1v1s
- 8 TECs and 10 Kapton heaters integrated
- 4.3 km elevation gain
- 2 chiller “incidents”
- And 1 TVAC chamber later…



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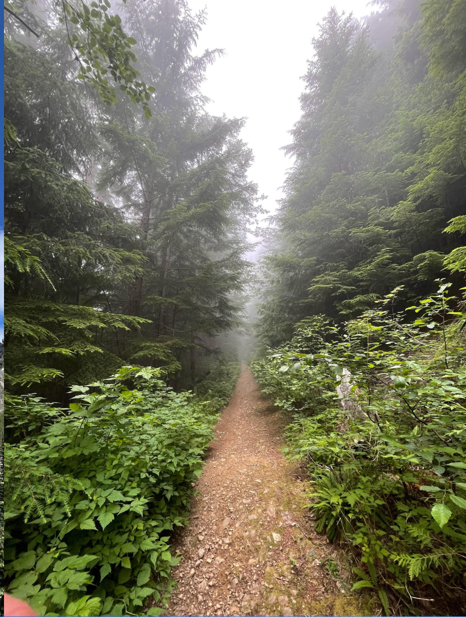





TVAC System Overview

Purpose

- Bring thermal vacuum qualification and acceptance testing in house for some Starfish payloads to save time and money

Goal

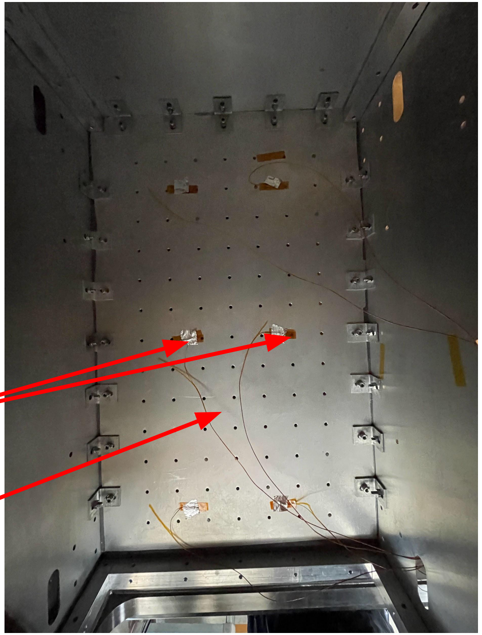
- Execute and log temperature profiles between - 35°C and + 100°C

Size

- Footprint: 457.2 mm x 711.2 mm
- Height: 430 mm

Construction

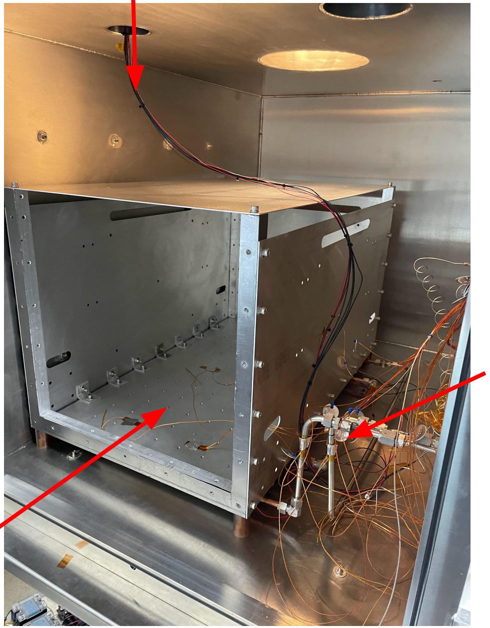
- Al-6061 breadboard, panels
- Al-6063 vertical posts
- Ultem standoffs/feet



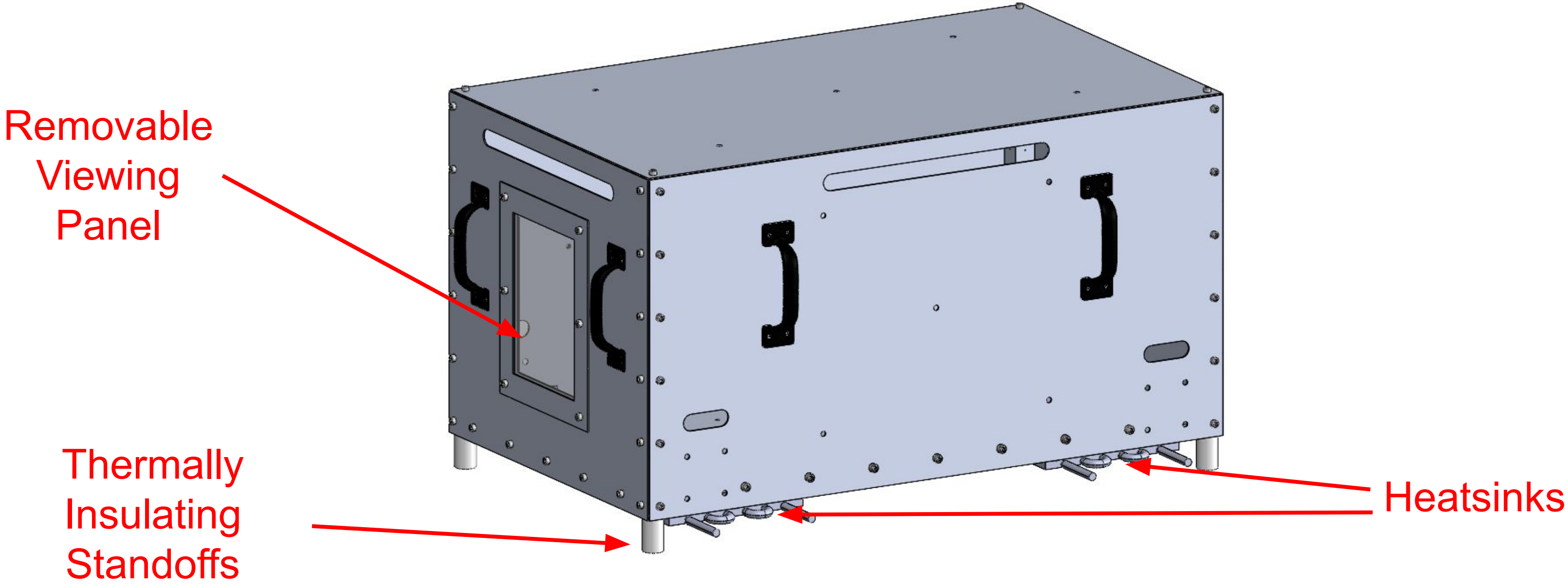
TVAC System Overview

Main systems and components

- Thermoelectric/Peltier coolers: Cold cycles
- Kapton heaters: Hot cycles
- Heatsinks: Improved ΔT
- Plumbing: Circulate chilled fluid
- Harnesses: Transmit power
- Panels/Enclosure: Minimizing radiative heat loss
- Mounting plate: Fixturing for test articles



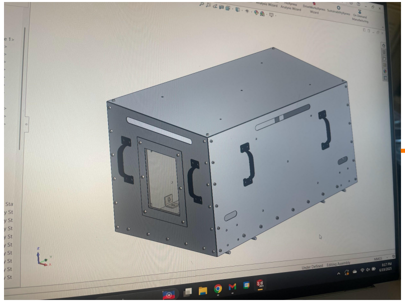
TVAC CAD Overview



CAD in current state

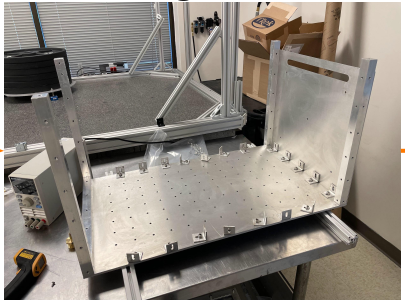
TVAC Evolution

Just a model

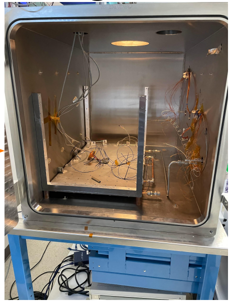


Chunks of metal

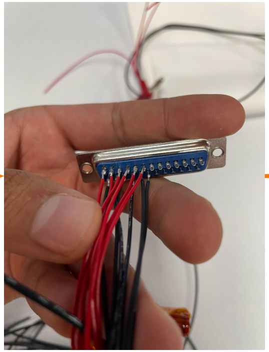
Mechanical Integration



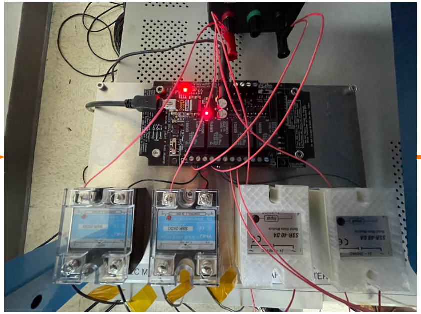
Thermal Integration



Harnessing



Power & Control



Testing

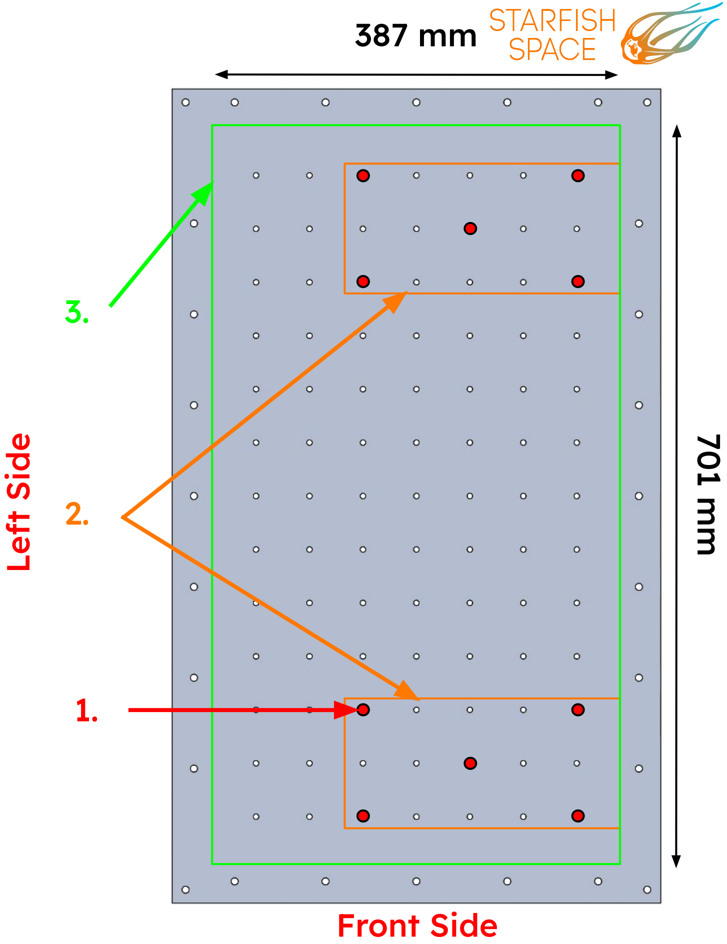
Running a Test

Mounting a test article

- There are 88 M6 holes available to connect your test fixture
- Holes spaced in a 7 x 14 array
- Distance is 50mm between holes

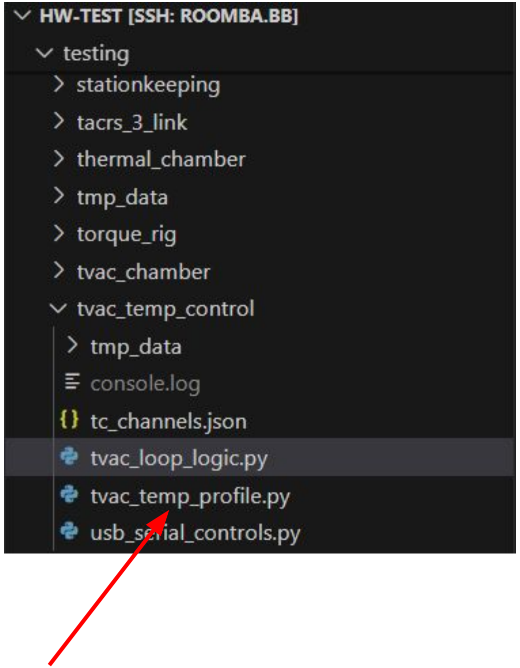
1: Holes are unavailable (used to clamp together the heatsink + TECs)
2: Screws should not extend > 0.25” past the top of the plate, or they will drive into the heatsink
3. Space for mounting

* Perspective from a top down view



Running a Test

- Integrate TVAC within vacuum chamber
- Mount test article
- Connect to Roomba.bb (to access TCs)
- Open a “screen” so you can run code without your laptop being on the same network
- Follow this path
  - hw-test > testing > tvac_temp_control > tvac_temp_profile.py
- Run script and input desired temperature profile in the form of a numpy array
- Ex. [[-20, 240], [-10, 240], [90, 60]] ramps to -20°C for 4 hours, -10°C for 4 hours, and 90°C for 1 hour
  - [[temp (°C), time (min)]]

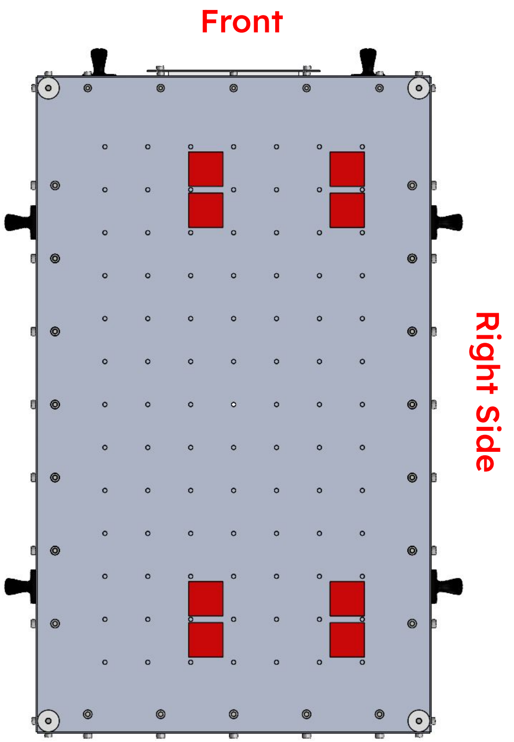


Cooling

TECs: Solid state heat pump that transfers heat across itself to create hot and cold sides

- Easier to handle and control over alternatives like cryogens
- Cooling capacity and heat dissipation are temp. difference between the hot and cold sides
  - ↑ ΔT ↓ Q_c
- Module selected for relatively high cooling power at selected operating current, and ability to cycle on/off without damage

* Perspective from a bottom up view
** TECs are off center to make plumbing more accessible



Cooling

Temperature Range

- -30°C cold plate temperature (lowest I’ve tested)

Heatsinking

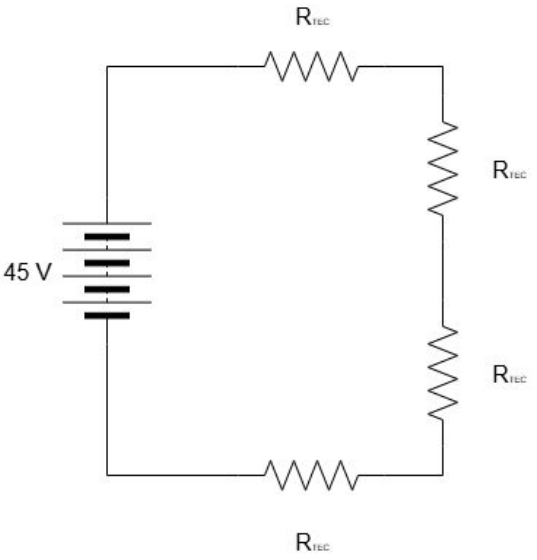
- Clamped between the aluminum cold plate and aluminum heatsinks (225 psi)
- Silver-filled thermally conductive vacuum grease on both faces

Wiring

- Two rails with four TECs on each
- Draw < 6A at 45V ; R_{TEC} is temp. Dependent
- Correspond to channel 1 on relays

Placement

- Mounted between the aluminum plates on the front and back



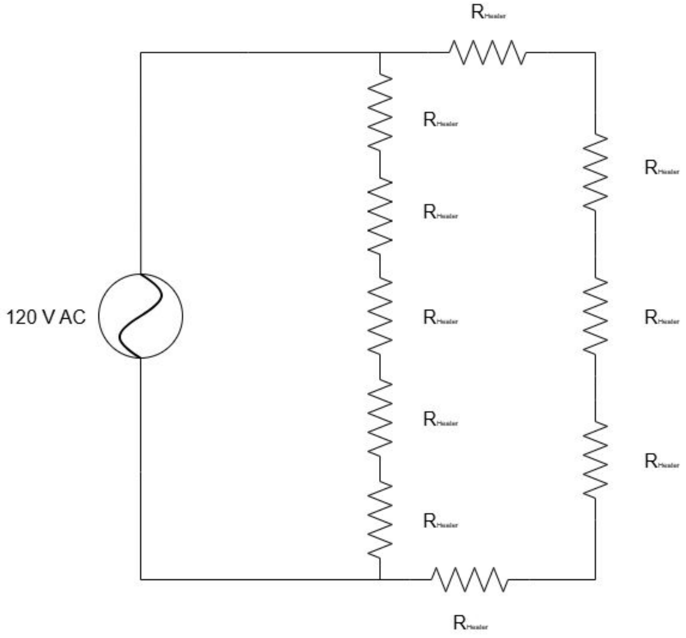
Heating

Kapton Heaters

- Generate heat via the joule effect
- ↑ Current ↑ Heat generation
- Operable with AC/DC current
  - We are running 120 VAC from the wall using the Variac transformer
- Mounted using a pressure sensitive adhesive
- Q_h = 50 W / heater

Wiring

- Placed on two rails to draw lower current and use existing wire (too much amperage means we would've needed lower gauge wires)
- Correspond to channel 2 on relays



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Heating

Wiring

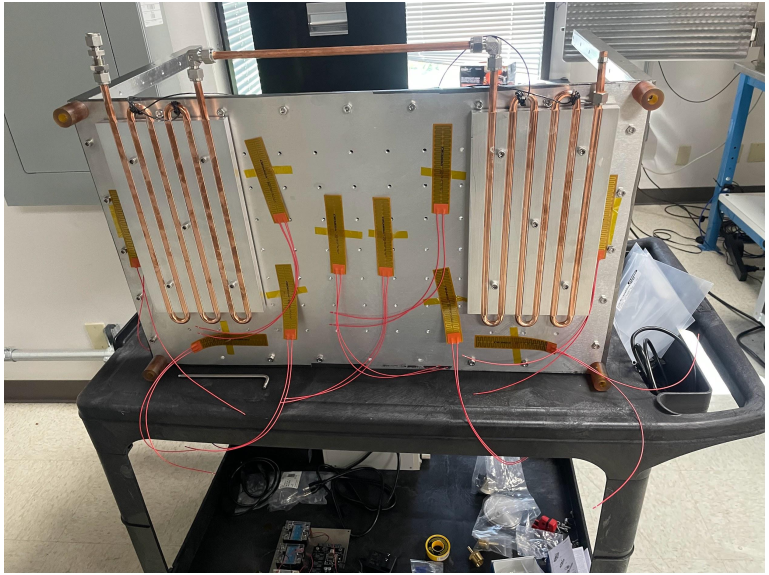
- Placed on two rails in parallel to draw lower current and use existing wire (too much amperage means we would've needed lower gauge wires)

Placement

- 6 heaters are located centrally
- 4 heaters split between the front and back

$R = V^2 / P = 28^2 / 50 = 15.68\ \Omega$

$I_{max} = \sqrt{(P / R)} = \sqrt{(50 / 15.68)} = 1.8\ A$



The heaters are in roughly this configuration (but stuck on and wired together)

Thermal Challenges

Temperature gradients

- Thermal interface material between parts to improve contact
- ΔT of 18°C between coldest point (directly above TECs and warmest point (center of top panel)
- Did not use MLI because these temperature gradients were not extraordinary and we did not have enough material to cover the entire chamber

Thermal runaway

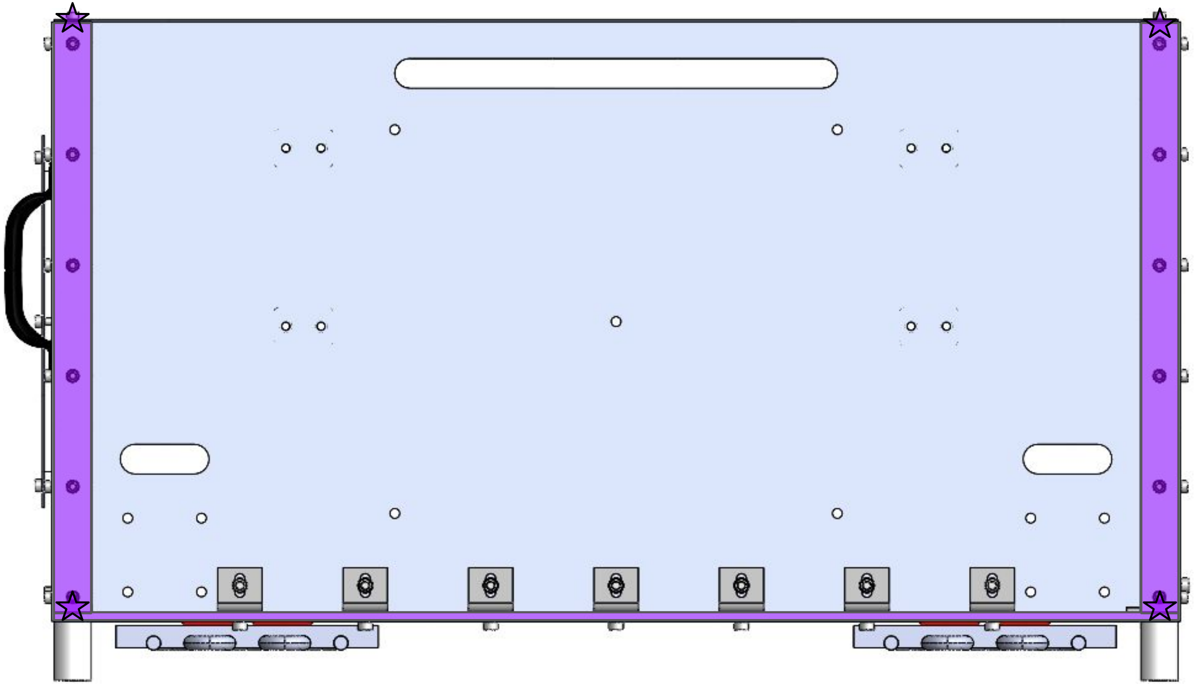
- At maximum expected current, the heat dissipation exceeded the chiller's cooling capacity, causing the system to warm up
- Running at 11.5 V instead

Temp. at cold plate

Temp. at center of top panel

Thermal Interfacing

Graphite TIM applied on all contact areas (indicated by purple box or star)



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Electronics

Harnessing

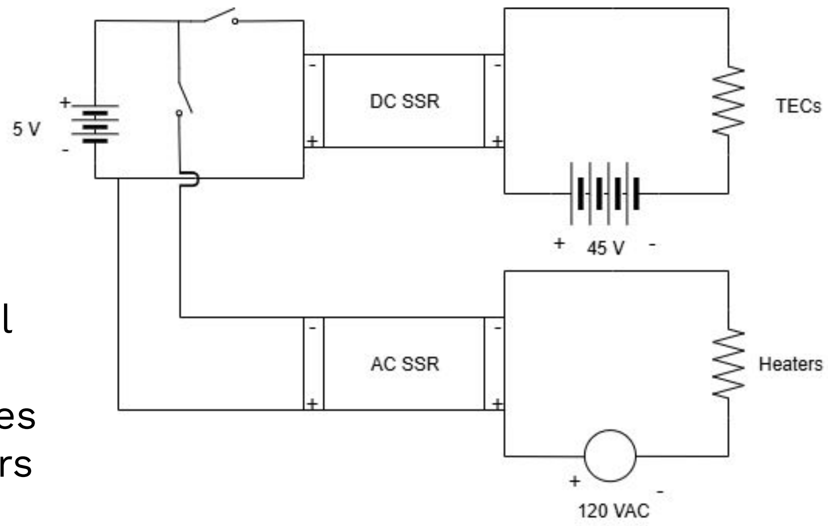
- Power is fed through a DB25 connector
    - Each pin is rated for 3-5A
- 18 AWG split into three parallel branches to accommodate ≤ 8A

Relays

- Electromechanical relays controlled by USB serial commands (low current/voltage) control voltage to solid state relays
- Solid state relays interface with power supplies and are directly connected to the TECs/heaters

Power

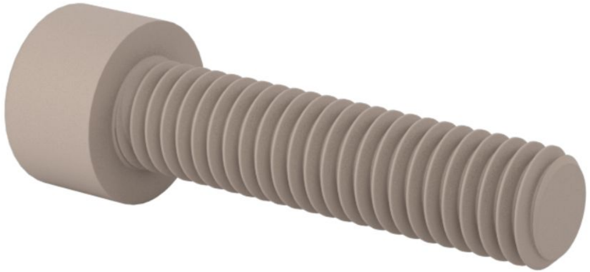
- TECs run on 45V and ≤ 6A
- Heaters run on 120 VAC and ≤ 2.5A



Troubleshooting pt. 1

Thermal contact

- TECs experienced lower current draw than expected indicating poor contact between the faces and the aluminum plates
- $\tau = (K \times D_{screw} \times P_{clamping} \times A_{TEC}) / N_{screws} = 13.4$ lb-in
  - Helpful calculator
- Providing more even clamping pressure across the TECs (using a torque screwdriver) and replacing steel screws with PEEK helped mitigate these effects
- At $T_{control} = -20^{o}C$; $T_{ambient} = -5^{o}C$, I_{drawn} increased from 4.6A → 5.3 A (at 45V)



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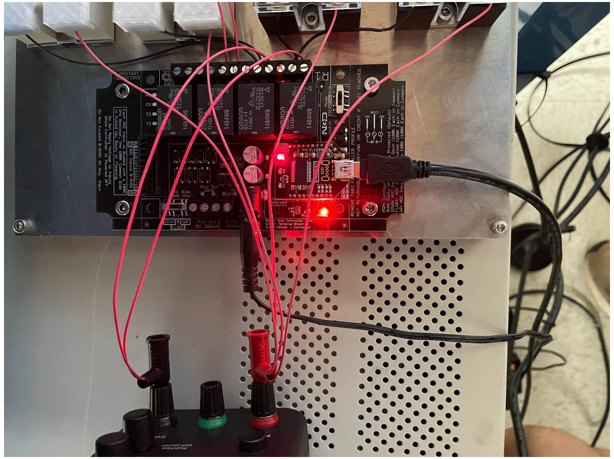
Troubleshooting pt. 2

Serial Commands

- Serial command documentation from this booklet
- Sending commands to turn off a single relay worked fine, but commanding multiple relays caused some issues
  - For example, a consecutive command to turn on channel 1 and turn off channel 3 turned on channels 1-4…
- The solution was to send commands with a small delay between them – time.delay(0.05)
  - Likely due to interference between commands since they were sent as USB serial bits



Troubleshooting pt. 3

Chiller Integration

- During a temp. profile that includes hot and cold temps., the circulating fluid presented a challenge
- Originally, the plan was to introduce a parallel path for the fluid to be diverted when cooling was not required
  - The solenoids and fittings massively decreased the flow rate
- Solution?
  - Accidentally discover a shut off feature on the back of the back of the chiller
  - Controlled by relay 3 (connecting the pins turns the chiller off)

Even this small system had slow fluid velocity

Control Logic

Hysteresis = ± 0.3°C
Settling band = ± 1.25°C
Cold Side

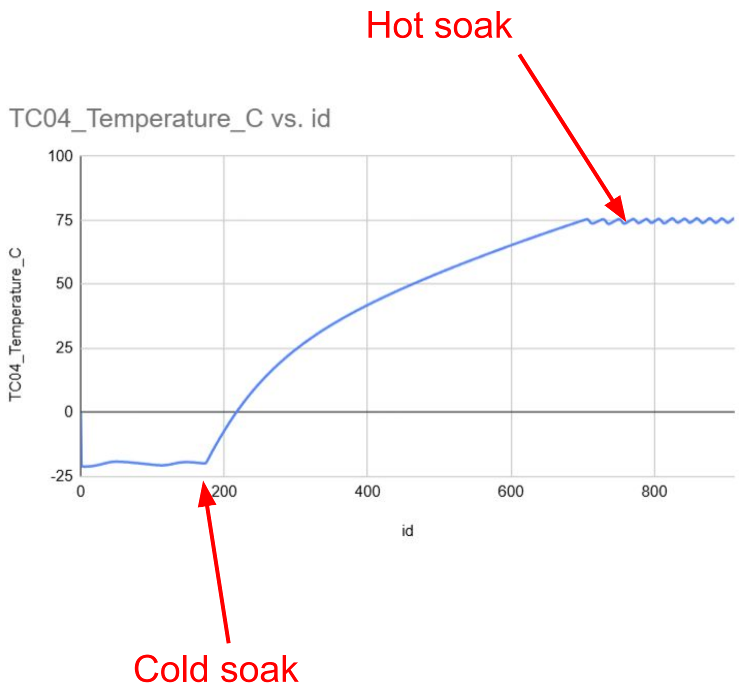
- TECs are running when the set temp. < 23.4°C (below ambient)
- If the measured temp. is below (T_{set} - 0.3°C), the TECs switch off

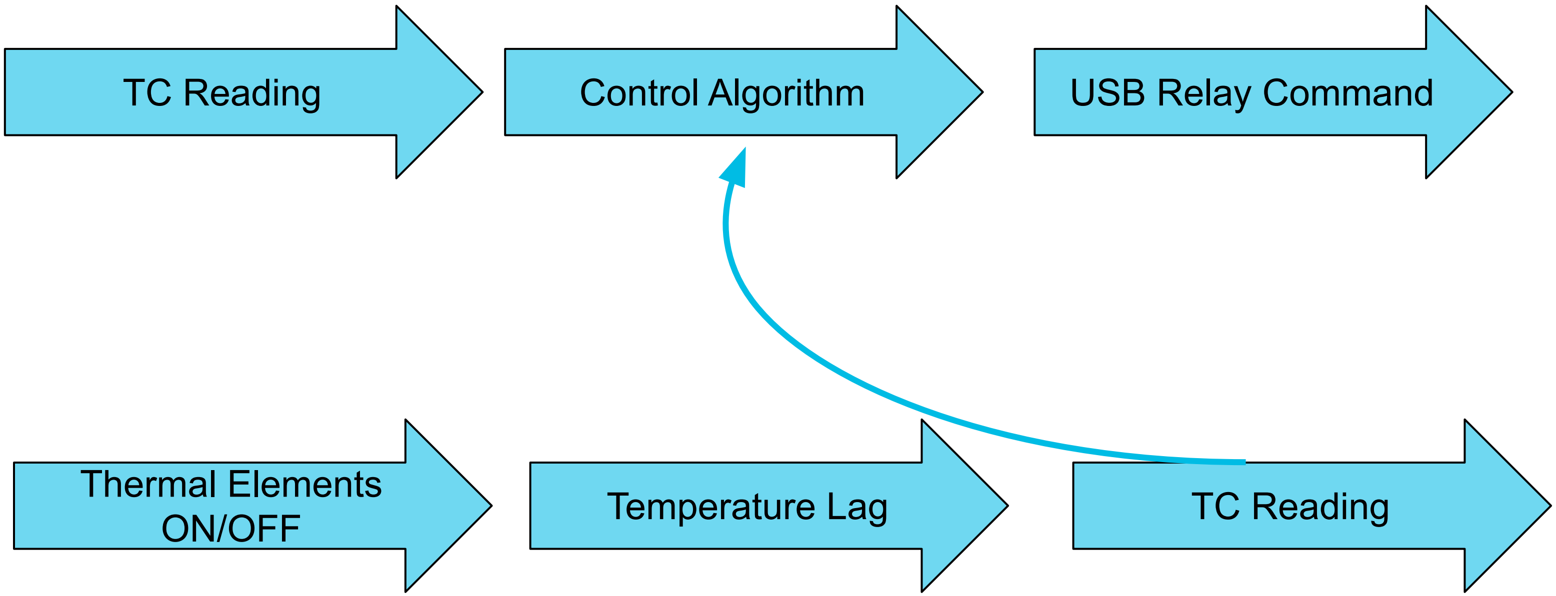
Hot Side

- Heaters are running when the set temp > 23.4°C (above ambient)
- If the measured temp. is above (T_{set} + 0.3°C), the heaters switch off

Soak

- When the measured temp. is within a settling band of the set temp., the soak period is initiated




TC Reading
Control Algorithm
USB Relay Command
Thermal Elements ON/OFF
Temperature Lag
TC Reading

Technical Hurdles

Software

- First I installed the wrong OS, then the wrong version of SolidWorks...

Chiller

- The first chiller didn't work – thankfully the replacement did (thank you ebay seller!)

Code

- Figuring out how to interface the DAQ, serial relays, BB, and control loop
- Debugging serial communication

Heat dissipation

- The TECs dissipate > 1 kW of heat

Resistance and contact issues

- Thermal paste isn't magic

yay

Takeaways/What I learned

- It is important to understand the limitations of your initial calculations based on the assumptions you baked into them
  - This might mean changing the scope of a project after some time
  - In our case, perfect thermal contact, ideal thermal resistance, etc.
- Some parts of a project move slower than others, but it is important to recognize which parts those should be
  - Making smarter, well planned decisions now can save you time and headaches later
- Translating a thermal system from design to product present exciting and difficult challenges

Original scope

Overview

Needs and Summary

- Purpose
  - Allow for thermal testing in an existing vacuum chamber
- Temperature Range
  - **- 50C to + 100C**
  - Cooled by thermoelectric coolers (TECs) and heated by kapton heaters
  - Monitored using thermocouples
- Size
  - 609mm x 609mm x 400mm (l x w x h)
  - **M = 34.2 kg**

1 month plan

Date	Meeting Plans	Tasks	Goals
Pre-Start	n/a	[Optional] Read through background material	Get excited for your first week!
June 2	Onboarding Meetings, Nautilus Weekly Planning	Intro meetings, onboarding setup tasks, access to all required resources	E-mail, Google Chat, Solidworks, Prusa Slicer,etc. working
June 3	Daily tag-up, 2x one-on-ones	TVAC part 1) Thermal Vacuum system design	Define the project, identify and research cooling options
June 4	Daily tag-up, 2x one-on-ones	TVAC part 1) Thermal Vacuum system design	
June 5	Daily tag-up, 2x one-on-ones	TVAC part 1) Thermal Vacuum system design	
June 6	Daily tag-up, one-on-one with intern lead (Sean)	TVAC part 1) Thermal Vacuum system design	Complete V1 of the thermal vacuum cooling system design
June 9	Weekly Planning, Company wide all hands meeting	TVAC part 1) Thermal Vacuum system design	mini-design review and make revisions
June 10	Daily tag-up, 2x one-on-ones	TVAC part 1) Thermal Vacuum system design	Order all parts
June 11	Daily tag-up, 2x one-on-ones	TVAC part 2) Build temperature/pressure monitoring system	
June 12	Daily tag-up, 2x one-on-ones	TVAC part 2) Build temperature/pressure monitoring system	
June 13	Daily tag-up, 2x one-on-ones, one-on-one with intern lead (Sean)	TVAC part 2) Build temperature/pressure monitoring system	integrate pressure gauge/thermocouple readings into a single setup
June 16	Weekly Planning	TVAC part 2) Build temperature/pressure monitoring system	
June 17	Daily tag-up	TVAC part 2) Build temperature/pressure monitoring system	software v1 ready for testing
June 18	Daily tag-up	TVAC part 3) Write software for automated temperature controller	Add automated temperature control to the software
June 19	Daily tag-up,	**Company holiday- take the day off**	
June 20	Daily tag-up, one-on-one with intern lead (Sean)	TVAC part 3) Write software for automated temperature controller	
June 23	Weekly Planning	TVAC part 3) Write software for automated temperature controller	software v2 ready for testing
June 24	Daily tag-up	TVAC part 3) Write software for automated temperature controller	
June 25	Daily tag-up	TVAC part 4) Begin thermal vacuum system build	All parts received?
June 26	Daily tag-up	TVAC part 4) Begin thermal vacuum system build	
June 27	Daily tag-up, Four Week Check-in	TVAC part 4) Begin thermal vacuum system build	
Future			
Last two weeks	Prepare final presentations		

What I am Leaving Behind

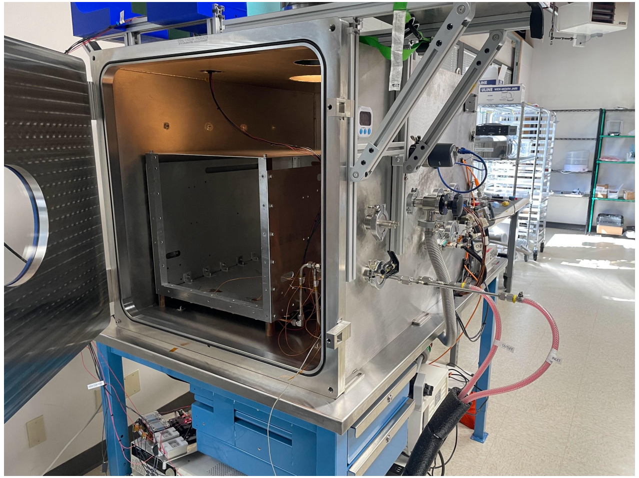
Documentation

- Includes design justifications, thermal calculations, component sizing calculations, CAD, part manuals, ordering sheets, etc.
- Split between Confluence and Google Drive

User Guides

- Google slideshow
  - Demonstrates procedure for integrating and de-integrating TVAC
  - Instructions for running a test

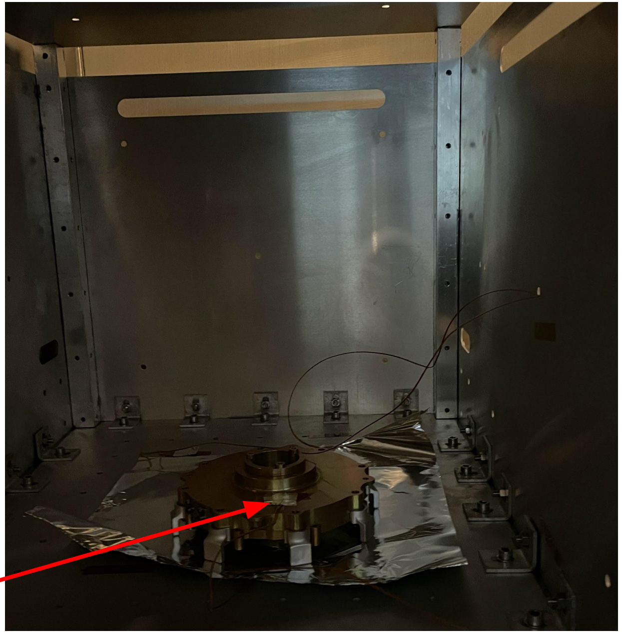
* I walked Lee and Arthur through what a test might look like (from the point of the TVAC being inside the chamber)



Next Steps for TVAC

- Chiller with a higher cooling capacity, or finding other ways to bring down the base temp of the heatsinks
- Improved temperature control loop (PID?)
  - Right now, the control algorithm is pretty simple (bang-bang)
  - Accounting for slower thermal lag if the TC being read is away from the thermal plate
- Continued testing with Starfish payloads
  - We are currently testing the Nautilus 2d damper for 24B!

TC04



The thermal contact isn't great unfortunately

Thanks!
Questions?

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What to include...

- Who I am
- Project
  - Original Scope vs. Today
  - Technical Hurdles
  - Technical skills acquired
  - Lessons learned
  - Thermal stuff
    - TEC PRO!
  - Harnessing
  - Troubleshooting
  - Coding stuff
  - Beagel bones
  - 1 on 1s
  - Wrapped stats
  - Seattle pics